

SURV/SURVMETH 720

Measurement Error Overview

Class #7

October 27, 2025

Midterm

- Grades posted
- Thoughts ?

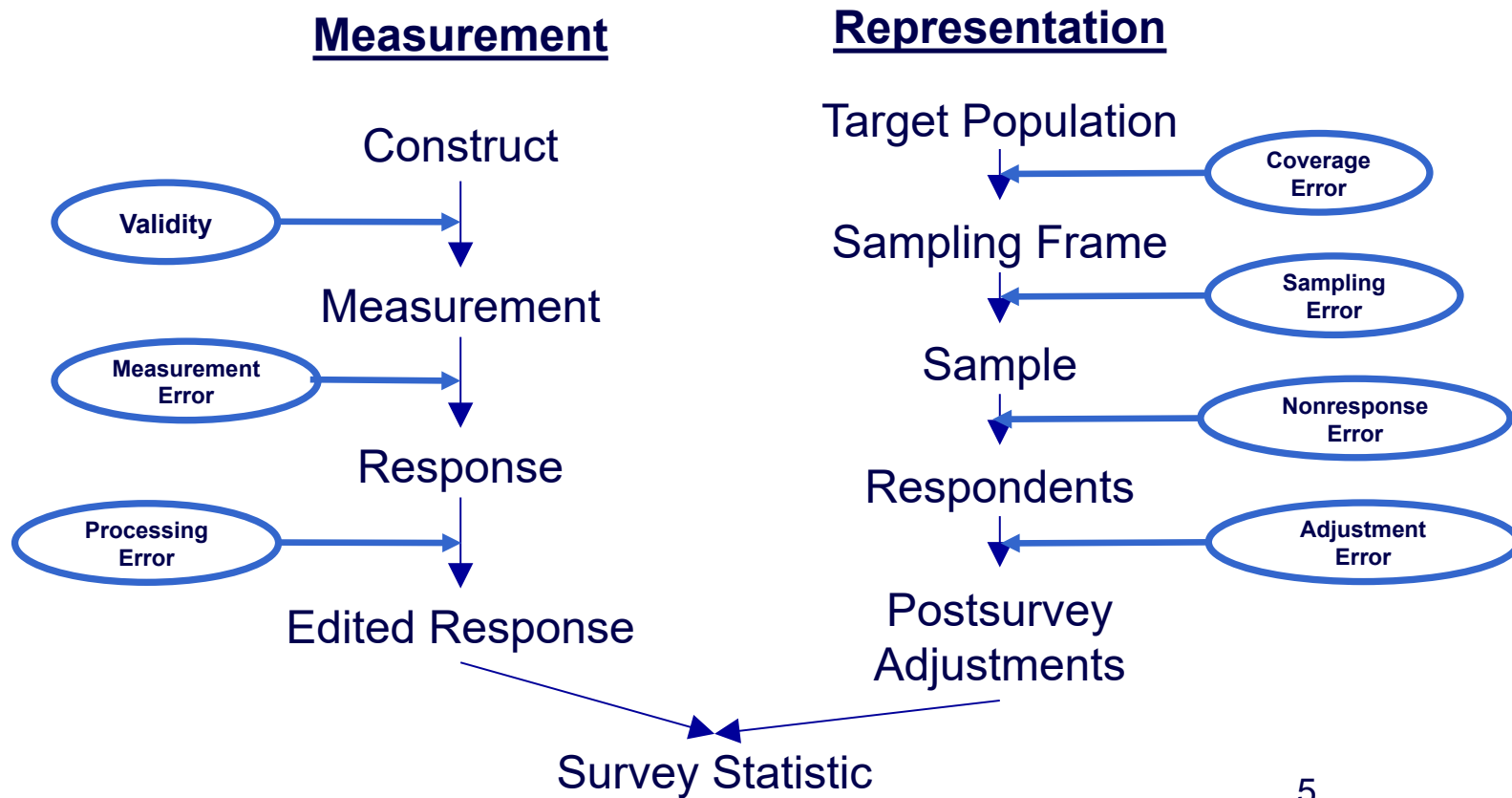
How about doing the midterm
again as a group ?
(ok, just one of the questions)

https://bit.ly/720_2025_Class7

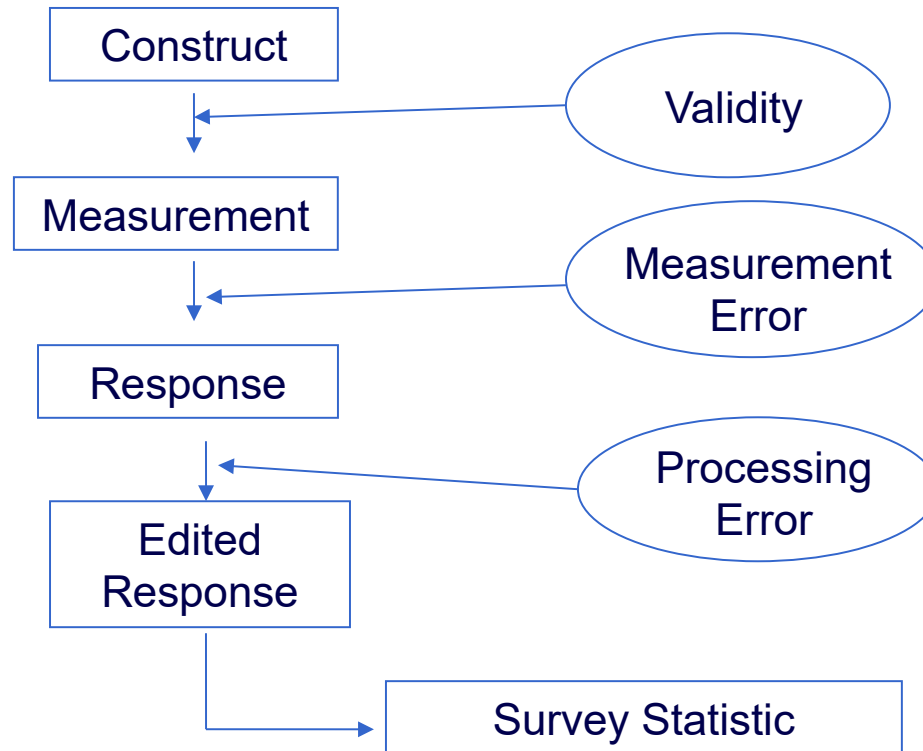
Today's Agenda

- Midterm review
- Construct validity overview
- Measurement error overview

TSE Diagram



TSE: Measurement



Construct validity

From Construct to Measurement

- Construct/Concept: Elements of information sought by researcher
 - Usually described by words, often abstract, permitting a definition
 - Examples: “job satisfaction”, “reading ability”, “religious attitudes”, “happiness”, “quality of life”, “crimes with victims”
- Measurement: Linking theoretical constructs to observable variables – ways to gather information about constructs
 - Examples: questions, observations, soil samples, blood pressure readings
 - Some constructs consist of just one attribute (e.g., age)
 - Other constructs are multi-dimensional with several attributes and need more than one Q to fulfill a particular objective (e.g., SES, attitudes)
 - Usually, numerous (competing) ways of how to measure a construct

- “Historically the most common definition of validity is that it refers to the extent to which a test or a set of operations measures what it is supposed to measure.”
- Ghiselli et al. (1981)
- Construct validity refers to how well you translated or transformed a concept, idea, or behaviour that is a construct into a functioning and operating reality, the operationalization. – Taherdoost, 2016
- How useful/appropriate are the answers to my question in telling me about what I am trying to measure ?

Construct Validity in Surveys

- Goal of question development is to choose wording that can successfully operationalize construct
- Validity is about extent to which each question measures the corresponding construct
 - Is the measure appropriate (measuring what it's intended to measure) and comprehensive (covering all aspects of the construct)?
 - Messick (1989): “even for purposes of applied decision making, reliance on criterion validity or content coverage is not enough. The meaning of the measure, and hence its construct validity, must always be pursued”.
- Mismatch between the information requested and the corresponding construct will produce invalid measures

Validity Examples

- Goal is to measure total income before taxes
 - Income can be reported before or after taxes, so probably not enough to ask “What is your total income?”
 - Q1: What was your total income before taxes?
- Goal is to measure job satisfaction
 - We assume job satisfaction depends at least on duties, compensation, environment, so probably not enough to ask “How satisfied are you with your job?”
 - Q1a: How satisfied are you with the work you do as part of your job?
 - Q1b: How satisfied are you with the amount of money you are paid?
 - Q1c: How satisfied are you with the facilities in which you work?

- How would you measure “be social”?

- How would you measure “socially active”?

- How would you measure “sociability”?

- How would you measure “be social”?
- How would you measure “socially active”?
- How would you measure “sociability”?

In sum, we can say that a not inconsiderable part of the precariousness of survey responses does not have anything at all to do with measuring instruments and measurement. It is present as a natural element in the reality of what is being measured – due to linguistic prerequisites and unclear ideas on the part of the individual.

Wikman, 2005

Assessing Validity

- **Qualitative** methods (e.g., cognitive interviewing; expert review): identify problems and reason for problems
 - Goal is not quantifying problems
- **Quantitative** methods ('empirical validity')
 - Validity of a measure is about correlation with the true score – sometimes have an observable true and sometimes not. If not:
 - identify the extent of problems and their statistical consequences
 - Correlation with another variable that on theoretical grounds should (or should not) be associated
 - Comparison between groups whose answers ought to differ if the answers are measuring the intended construct
 - Split-ballot type experiments

1. Sudman and Bradburn (1982)

Wording 1: On days when you drink alcohol, how many drinks do you usually have: would you say one, two, three, or more?

Wording 2: On days when you drink alcohol, how many drinks do you usually have: one or two, three or four, five or six, or seven or more?

% respondents reporting three or more drinks in
Wording 2 > Wording 1

2. Wood and Williams, 2007

- “The validity of self-reported gambling expenditure is questionable due to the fact that these expenditures usually do not match actual gambling revenue.”
- ‘How much do you spend on gambling in a typical month?’ Using the term ‘spend’ may bias people toward reporting losses rather than wins.
- Gerstein et al., 1999: asked whether they had ‘come out ahead or behind on your gambling’, with the choices being ‘ahead, behind, or broke even’. With this wording, a majority of people actually reported

Retrospective validity check: Experiment with 12 different question wordings.

Roughly how much money do/did you	spend	on gambling	in a typical month?	What we mean here is how much you are ahead or behind, or your net win or loss.
		in total on lottery, raffle and instant win tickets, Sports Select, slot machines and table games at Ontario casinos and racetracks, horse race betting, and bingo		
	come out ahead or behind	on specific gambling activity (8 different types)	last time you purchased/played (this activity)? How often do you purchase/play (this activity)?	

Figure 1 Segments of the 'retrospective gambling expenditure' question that were varied

- Criterion validity: authors performed a retrospective validity check:
 - Compare estimated expenditures, for each question, to actual gambling revenues in Ontario.
 - Also performed a prospective validity check via a diary

Median monthly expenditures produced by some questions were five times higher than those produced by other Qs !

E	Roughly how much money do you come out ahead or behind on gambling in a typical month?	-\$10.00
A	Roughly how much money do you spend on gambling in a typical month?	-\$20.00
D	Roughly how much money did you spend on [specific gambling activity] the last time you purchased/played [specific gambling activity]? Roughly how often do you purchase/play [specific gambling activity]? Would you say daily, 4–6 times a week, 1–3 times a month, 4–12 times a year, 1–3 times a year, or less than once a year?	-\$49.85

3. Tourangeau et al., 2018

- Population Assessment of Tobacco and Health (PATH) Study

Relatively few national surveys assess the reliability and validity (RV) of answers to survey questions, although most survey researchers agree that valid and reliable data are central goals for questionnaire designers.¹

- Two methods to assess validity:

- Saliva samples at the end (of the reinterview)

Survey data and saliva test results agree for 87.5% of the respondents. Discrepancy due to “occasional users” testing negative.

- Photograph any tobacco products respondents had reported.

At the product level, the photographs matched the survey reports 98.1% of the time.

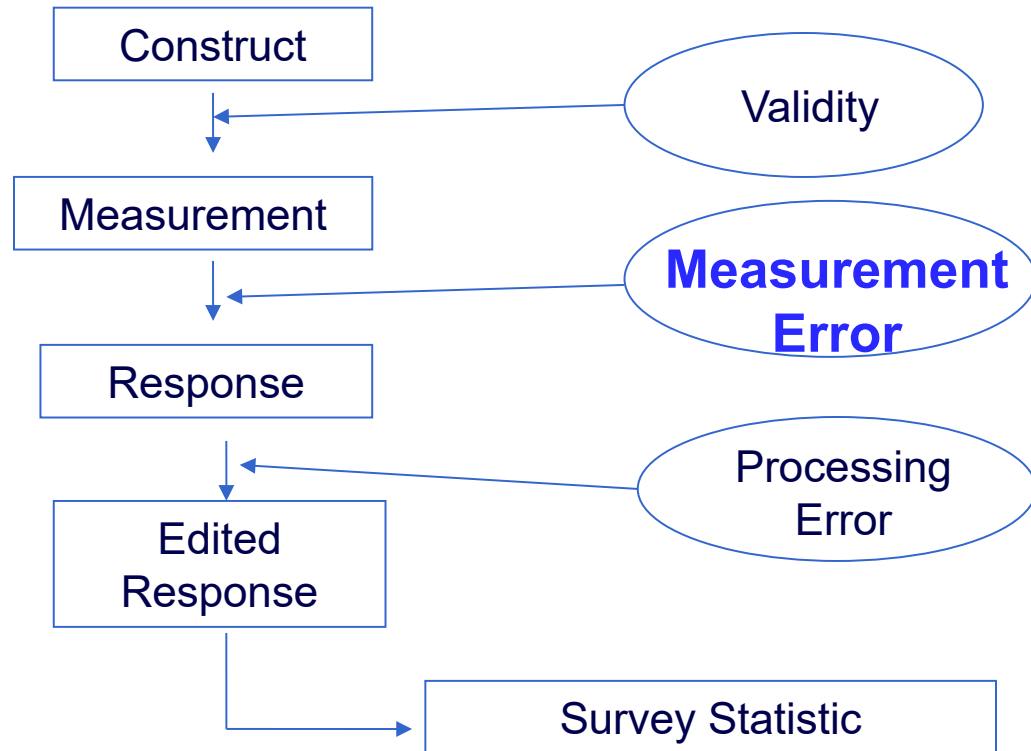
Incidentally...

“Prior to 1954 (seems like a long time ago) the concept of validity was in considerable disarray. There were almost as many definitions and varieties of validity as there were people interested in psychometric theory.”

– Ghiselli et al. 1981

Incidentally...

- "Validity" is a term used in somewhat different senses by different disciplines and, even within survey research, it seems to mean different things to different researchers.
 - Groves et al. (2004, p. 254)
- “The more one reads of errors in psychological measurement, the more types of validity that one encounters. Each author seems to feel both free to ignore phrases used in prior research and compelled to invent his/her own.”
 - Groves (1989, p.27)



Measurement Error

- Any deviations of the reported value from the “true” value
 - Can be systematic or random
- Has meaning at two levels
 - Respondent level
 - Estimate level
- Two ways of thinking about “truth”
 - Platonic true score
 - Classical true score

Respondent Level

- Focus is on i-th person's response deviations from her true value
- A single observation (y_i) is represented as the sum of two terms: a true value μ_i and an some measurement error, ε_i .

$$y_i = \mu_i + \varepsilon_i$$

- It follows that measurement error is the difference between their response and their true value

$$= y_i - \mu_i$$

Respondent Level...

- This can be further decomposed into conceptual replications of measurement, where answers (and response deviations) vary over trials.
- The expected observed value across trials ($\bar{y}_i$) is now represented as a true value μ_i and the expected value of error across trials for the respondent, $\bar{\varepsilon}_i$.

$$\bar{y}_i = \mu_i + \bar{\varepsilon}_i$$

- Respondent-level bias exists when ε 's have a nonzero expected value (over conceptual replications of the questioning)

Respondent Level...

- Simple response variance is defined as the variance of responses over repeated trials
 - Will be zero if respondent provides consistent response under same conditions over time ('reliability')
- Consistency does not imply accuracy
 - Asking the same question at two points in time could yield the same answer, but the answer may be far from the true value

Estimate Level

- Focus is on the average error across respondents
- Respondent mean is represented as the true respondent mean and mean error across respondents (r):

$$E(\bar{y}) = E\left(\frac{1}{r}\sum_{i=1}^r u_i + \frac{1}{r}\sum_{i=1}^r \varepsilon_i\right)$$

- Estimate is biased when $\bar{\varepsilon}$ has a nonzero expected value

Estimate Level...

- Random measurement error accounts for error variances across respondents
- If response deviations across respondents average to zero, does random measurement error have any statistical consequences?
 - For *variance of an estimate*, yes. The greater the meas. error variance, the larger the variance of y will be, other things being equal.
 - For *point estimates*, it depends. Means and proportions are not affected, but correlation and regression coefficients will be attenuated (Biemer and Trewin, 1997)

Effect of Random Meas. Error

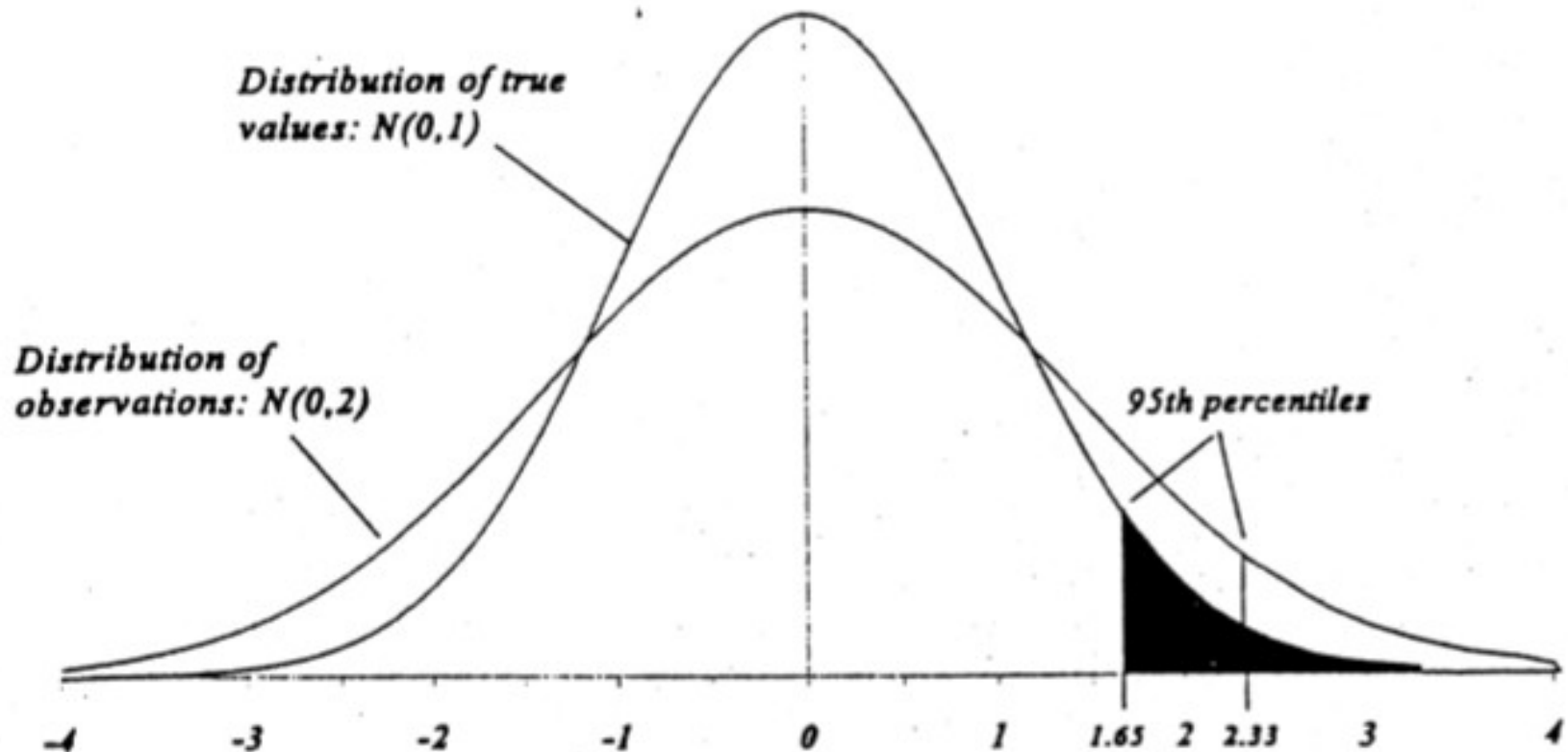


Figure 27.2 The effect of measurement error on the 95th percentile of an $N(0, 1)$ distribution.

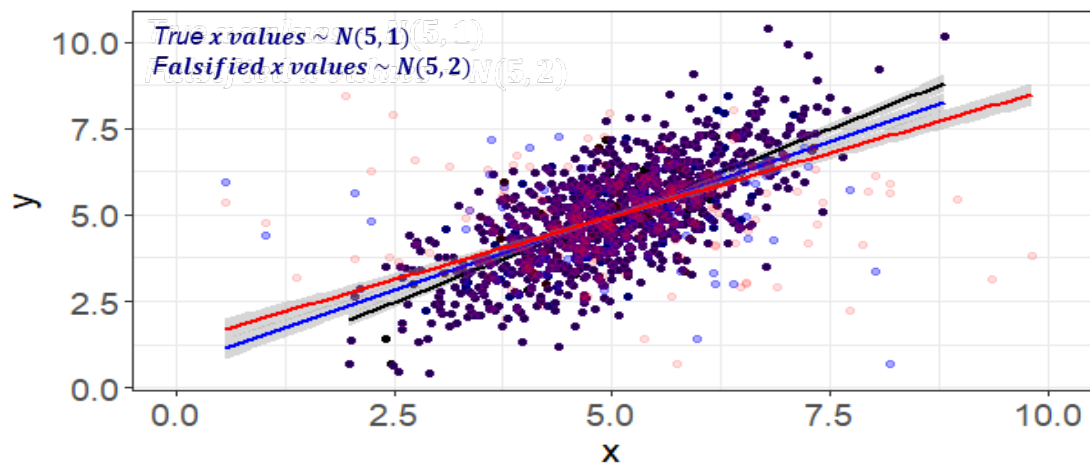
Biemer and Trewin (1997)

Simple linear regression

True regression
line

Regression line
(5% falsification)

Regression line
(10% falsification)

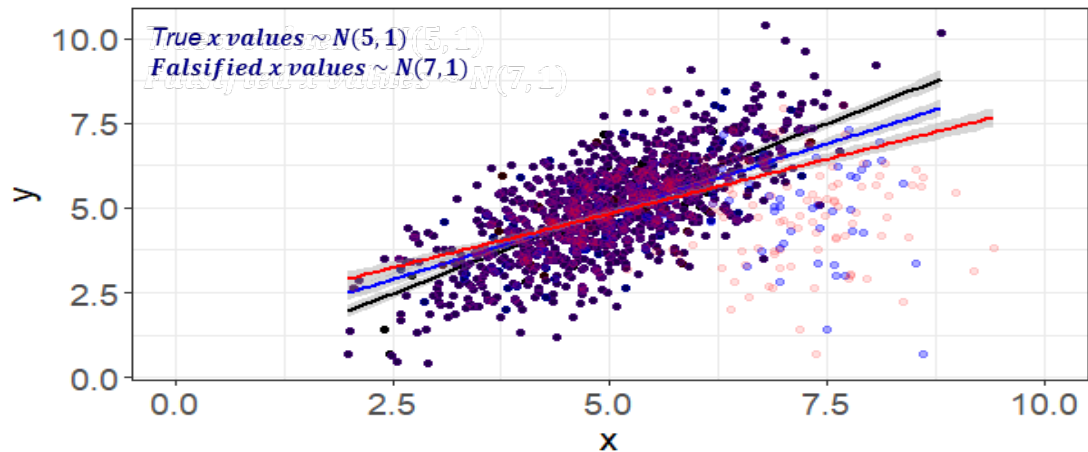
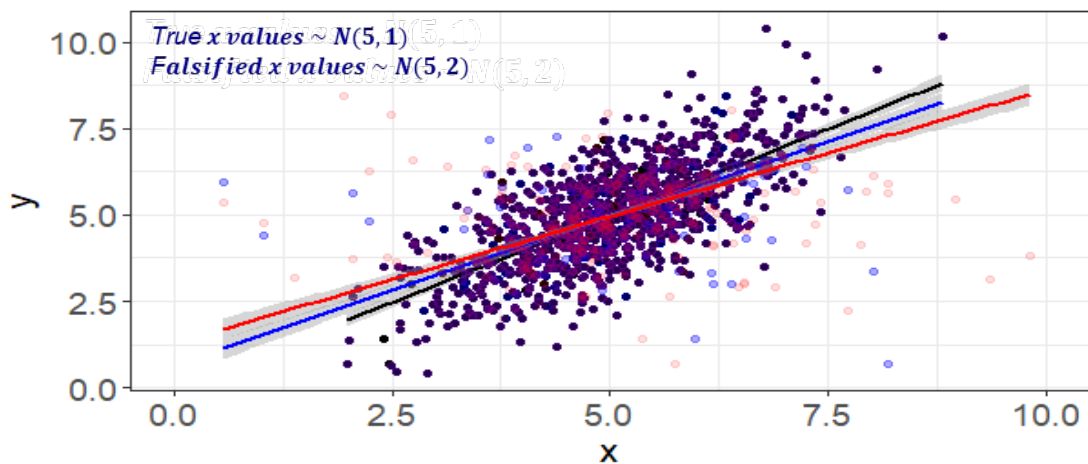


Simple linear regression

True regression
line

Regression line
(5% falsification)

Regression line
(10% falsification)



Falsified x values lead to biased
(attenuated)
coefficients.

$$\beta_1^{(F)} = \lambda \beta_1^{(T)}$$

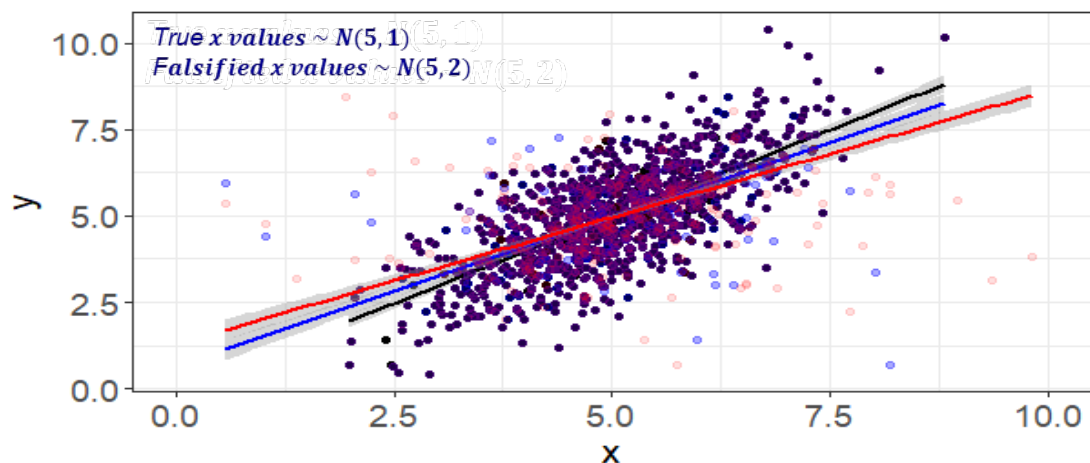
$$\lambda = \frac{\text{true variance}}{\text{total variance}}$$

Simple linear regression

True regression
line

Regression line
(5% falsification)

Regression line
(10% falsification)



What will happen to our inferences when there are measurement errors in 'y'?

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Measurement Error

- Any deviations of the reported value from the “true” value
 - Can be systematic or random
- Has meaning at two levels
 - Respondent level
 - Estimate level
- Two ways of thinking about “truth”
 - Platonic true score
 - Classical true score

Platonic True Score

- Construct being measured is at least, in theory, verifiable based on tangible non-survey evidence
 - In other words, “truth” can be derived from external source
 - Applies to behaviors and some individual characteristics
 - Examples: someone’s vote in 2024 election, household size, BMI
- Survey researchers largely adhere to this view

Classical True Score

- Construct being measured has no existing real-world measure
 - Applies to psychological traits and attitudes
 - Applies to latent constructs measured using multi-item scales
 - Examples: extraversion, job satisfaction
- True score is conceptualized as expected value if something were remeasured over a very large number of trials
 - “psychometric perspective”

Classical True Score Cont.

- At respondent level, a single observation (y_{it}) is the sum of two terms: the mean of the response distribution for an individual μ_i^* and an error for the t-th response from it, ε_{it} .

$$y_{it} = \mu_i^* + \varepsilon_{it}$$

Illustration: Classical True Score

- Suppose that on a particular measure of extraversion (0-100), sometimes the i -th respondent may achieve a 52 but on repeated administrations might achieve a 59, 49, and 56, respectively.
- What is their “true” level of extraversion according to the classical score theory?
- What are the response errors for each trial?

Illustration: Measurement Trials...

- True score is the mean of response distribution across trials
 $(52+59+49+56)/4 = 54$
- Errors are the deviation from the true score:

$$T1: 52-54 = -2$$

$$T2: 59-54 = 5$$

$$T3: 49-54 = -5$$

$$T4: 56-54 = 2$$

Assumption is random error across trials (this is not bias).

Next Week

- Review these concepts again + add some more concepts
- Methods of studying measurement error
- Reading assignment
- Questions?